

HONG KONG BAPTIST UNIVERSITY

COURSE OUTLINE

1. COURSE TITLE

Machine Learning

2. COURSE CODE

COMP7250

3. NO. OF UNITS

3 Units

4. OFFERING DEPARTMENT

Master of Science in Data Analytics and Artificial Intelligence

5. PREREQUISITES

COMP7015 Artificial Intelligence

COMP7180 Quantitative Methods for Data Analytics and Artificial Intelligence

6. MEDIUM OF INSTRUCTION

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7. AIMS & OBJECTIVES

To introduce the basic concepts, theories and techniques of machine learning from an application perspective. To give students practical insights into the current development of the field.

8. COURSE CONTENT

I. The Evolution of Machine Learning Algorithms and Applications

- Algorithm Level
- Application Level

II. Deep Learning from Application Perspective

- Models
- Learning Objectives
- Optimization and Regularizations
- Lab-1: ResNet Implementation
- Lab-2: LSTM Implementation

III. Foundation Models from Application Perspective

- Large Language Models
- Vision-language Models
- Learning Algorithms
- Deployment Strategies
- Lab-3: LoRA Implementation

IV. Other Topics in Modern Machine Learning

- Trustworthy Machine Learning for Science
- Lab-4: FGSM and PGD Implementation

9. COURSE INTENDED LEARNING OUTCOMES (CILOs)

CILO	By the end of the course, students should be able to:
CILO 1	Explain the capabilities, strengths and limitations of various machine learning techniques
CILO 2	Explain various machine learning algorithms and their applications
CILO 3	Describe learning models and algorithms
CILO 4	Apply machine learning algorithms to solve real-world problems, understand advanced machine learning algorithms, cultivate the ability to evaluate learning algorithms
CILO 5	Implement solutions to real-world problems using deep learning toolboxes such as PyTorch or Tensorflow

10. TEACHING & LEARNING ACTIVITIES (TLAs)

CILO alignment	Type of TLA
1-3	Students will learn the basic concepts and fundamental principles of machine learning, as well as the application examples, in lectures.
4-5	Students will work on assignments to enhance the understanding of learning principles, and acquire hands-on experience via laboratory classes and a mini project.

11. ASSESSMENT METHODS (AMs)

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Continuous Assessment	40 %	1-5	Assignments and a hands-on mini-project will be used to evaluate students' understanding of basic concepts and to assess their ability to apply

			learning theory to solve real-world problems
Examination	60 %	1-5	Examination will be used to assess students' overall understanding of various machine learning algorithms, their applications, as well as their capabilities, strengths and limitations.

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